

From hyperspectral CHRIS to four RGBN bands

The cloud model eats **4 bands** (R, G, B, NIR). CHRIS has **18 to 62 bands** per mode. So each pseudo-band is built from the CHRIS bands that fall in it.

Three ways to build them: the plain mean the model was trained on, then SRF-weighted resampling, then convolution with the CHRIS response.

METHOD 1 OF 3

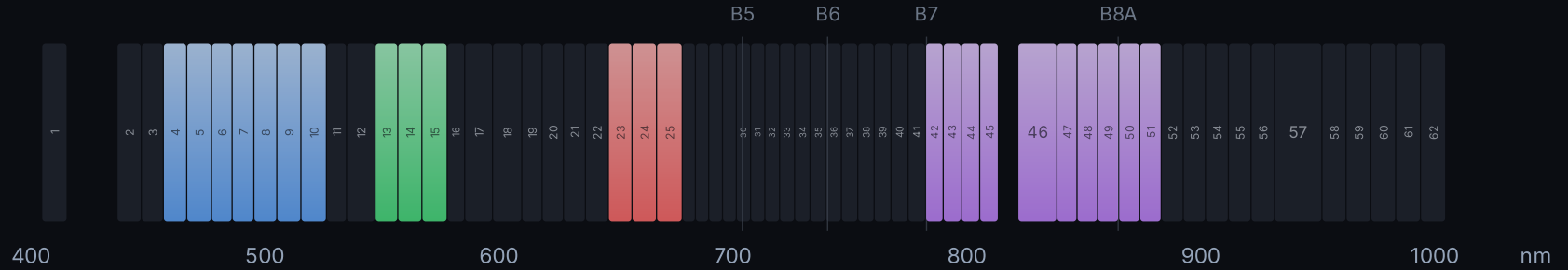
Plain mean

- which each CHRIS band goes to the Sentinel-2 band it overlaps the most
- how plain **mean** of the CHRIS bands assigned to each pseudo-band
- then DN/100 000 or TOA/10 000, clip to 5, and a per-tile z-score inside the network

$$\bar{L}_g = \frac{1}{N_g} \sum_{i \in g} L_i$$

This is the training mapping, the one the model actually learned.

Mode 1 : 62 bands, ~34 m



B2
459-525 nm



B3
542-578 nm

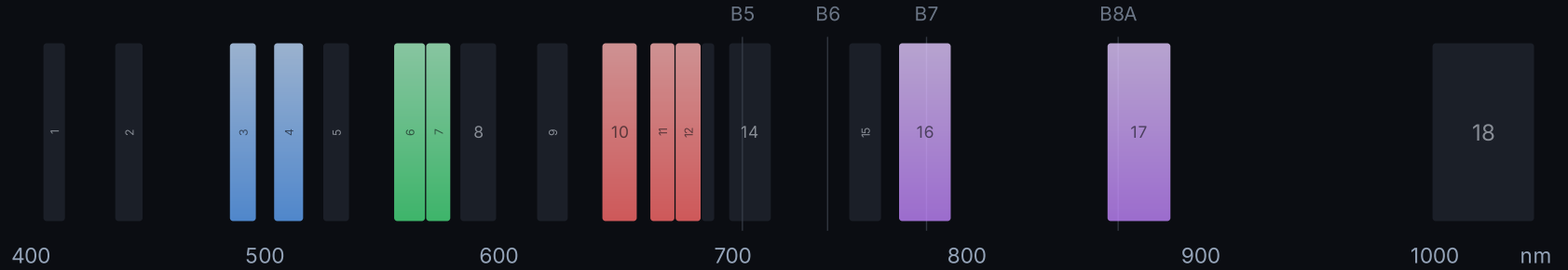


B4
649-680 nm



B8
780-886 nm

Mode 2 : 18 bands, ~17 m



B2

459-525 nm



B3

542-578 nm



B4

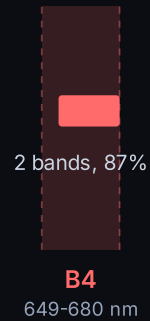
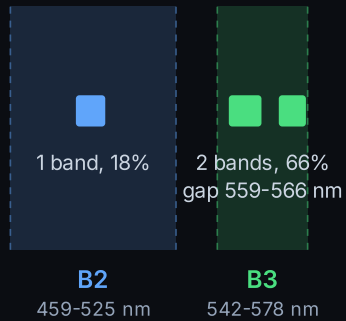
649-680 nm



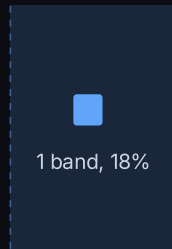
B8

780-886 nm

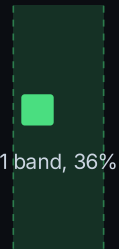
Mode 3 : 18 bands, ~13 m



Mode 4 : 18 bands, ~11 m



B2
459-525 nm



B3
542-578 nm

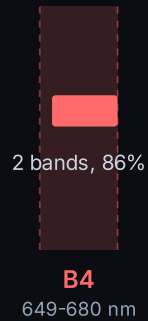
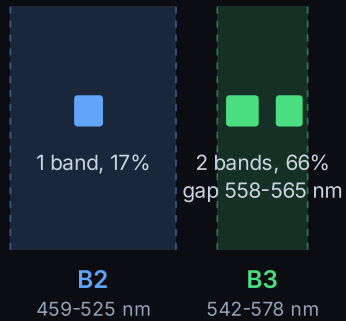
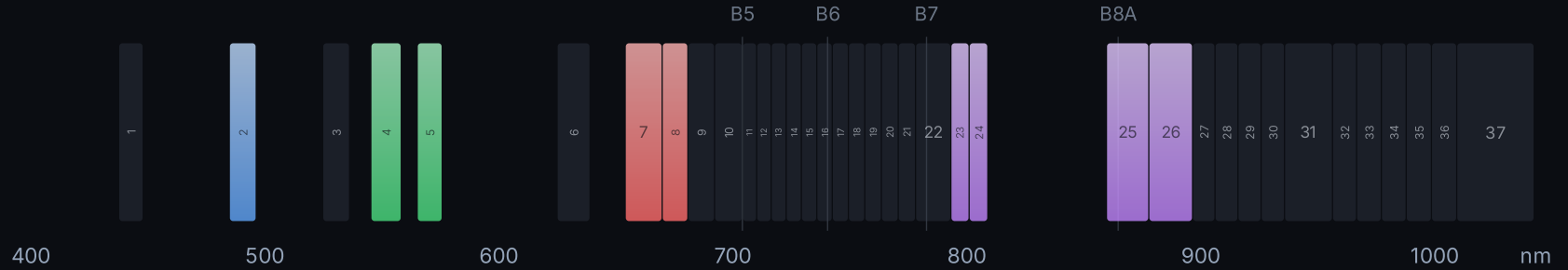


B4
649-680 nm

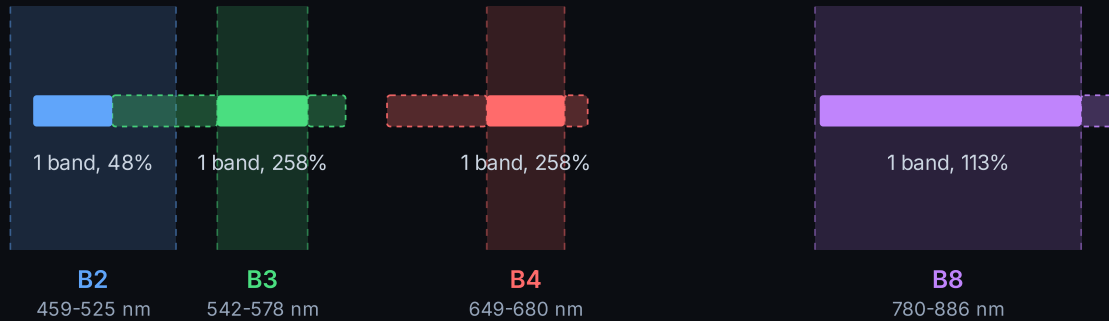


B8
780-886 nm

Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Method 1: where the bands end up

R, B4

G, B3

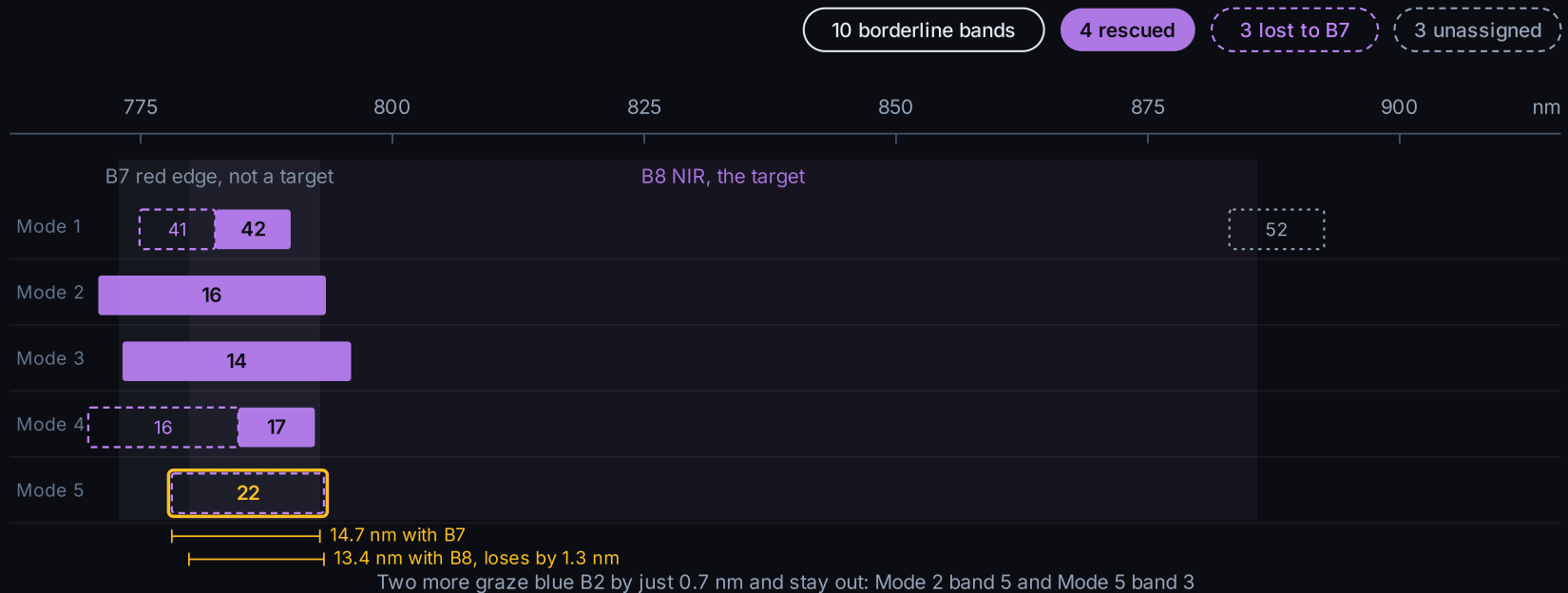
B, B2

NIR, B8

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	4-10 (7), 457-526 nm	13-15 (3), 547-578	23-25 (3), 647-678	42-51 (10), 782-883
2	3-4 (2), 485-517	6-7 (2), 555-580	10-12 (3), 644-686	16-17 (2), 771-887
3	2 (1), 486-497	4-5 (2), 546-577	7-8 (2), 656-683	14-15 (2), 773-890
4	1 (1), 484-496	2 (1), 545-558	4 (1), 664-675	17-18 (2), 785-800
5	2 (1), 485-496	4-5 (2), 545-576	7-8 (2), 654-681	23-26 (4), 793-896
20	1 (1), 469-500	2 (1), 500-593	3 (1), 609-689	4 (1), 782-901

Training mapping. Numbers are 1-based CHRIS band indices; nm is the spectral span. Stack order: R, G, B, NIR.

What the max overlap rule leaves out



Each of these bands touches the NIR target but the winner takes all rule sends it away. Four were rescued by hand for training, three were left out after losing the red edge B7 duel (including Mode 5 band 22, the heaviest NIR band the SRF sees), and three stay unassigned under the 30 percent overlap threshold.

METHOD 2 OF 3

SRF-weighted resampling

which every CHRIS band with nonzero S2A response takes part, no winner takes all

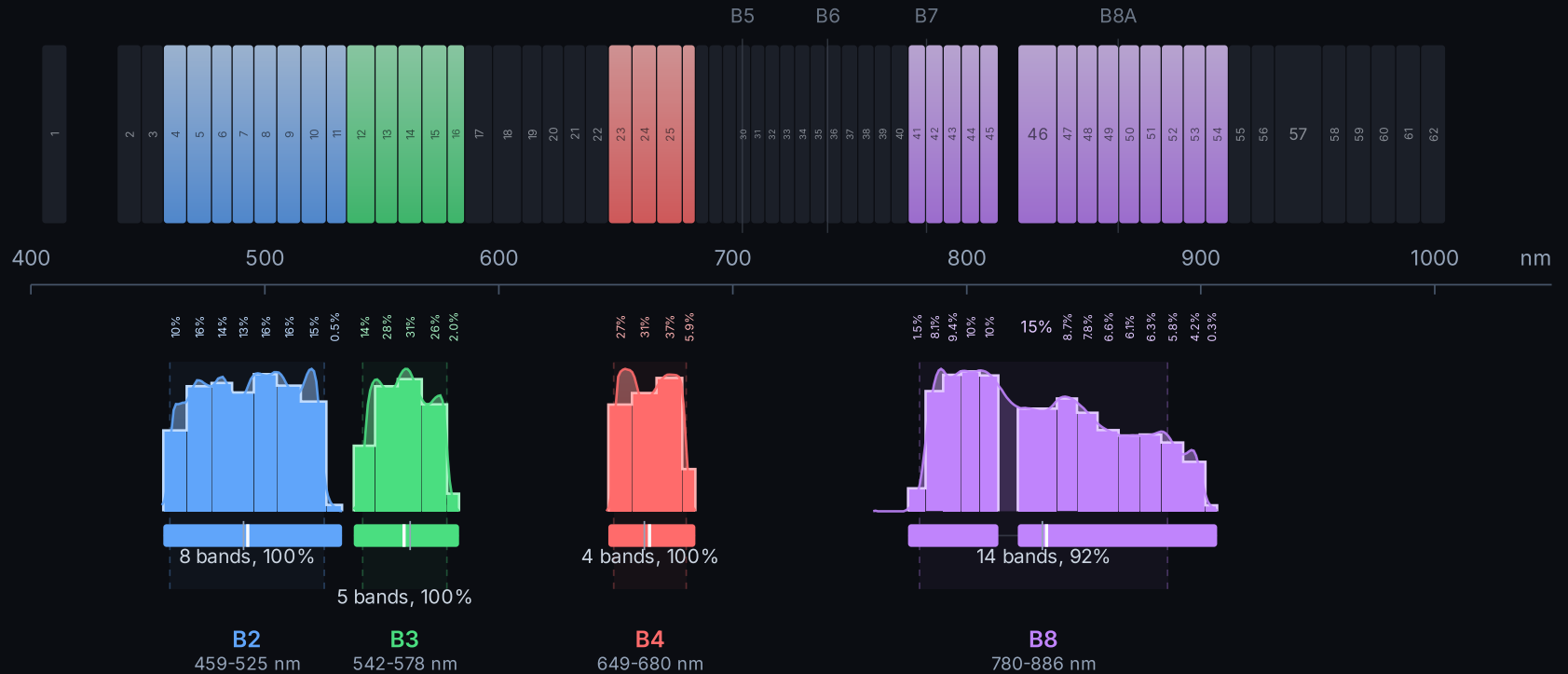
how w_i is the integral of the S2A response curve over the CHRIS band, and the pseudo-band is the weighted mean

then weights are normalized by the area CHRIS actually covers, not by the full SRF area

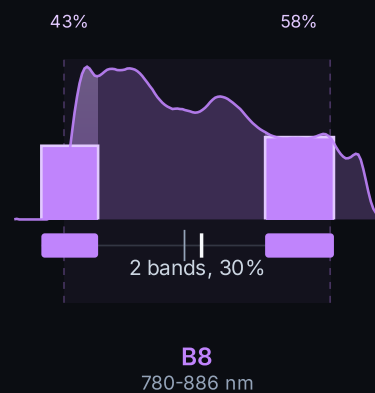
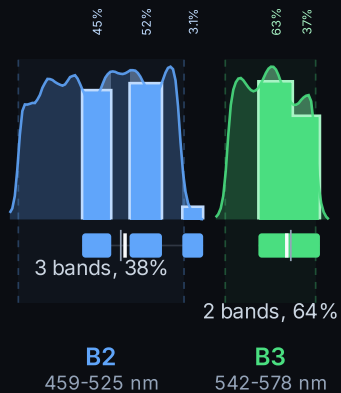
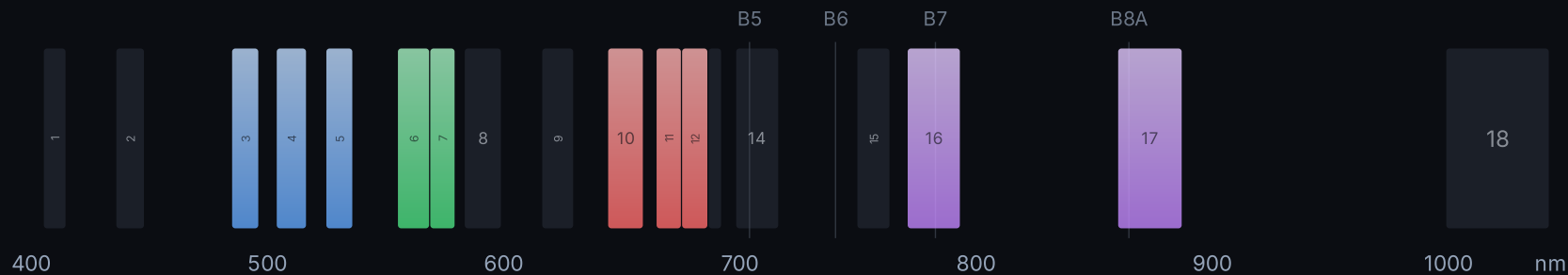
$$\tilde{L}_g = \frac{\sum_i w_i L_i}{\sum_i w_i}, \quad w_i = \int_{\lambda_i^{lo}}^{\lambda_i^{hi}} R_g(\lambda) d\lambda$$

Same four targets, read through the real S2A response curves instead of flat FWHM boxes. In the lanes: bands entering and the share of the response they cover; the white tick is the weighted centroid, the gray one the plain mean. The percentages above the lanes are each band's share of the mix.

Mode 1 : 62 bands, ~34 m



Mode 2 : 18 bands, ~17 m



Mode 3 : 18 bands, ~13 m



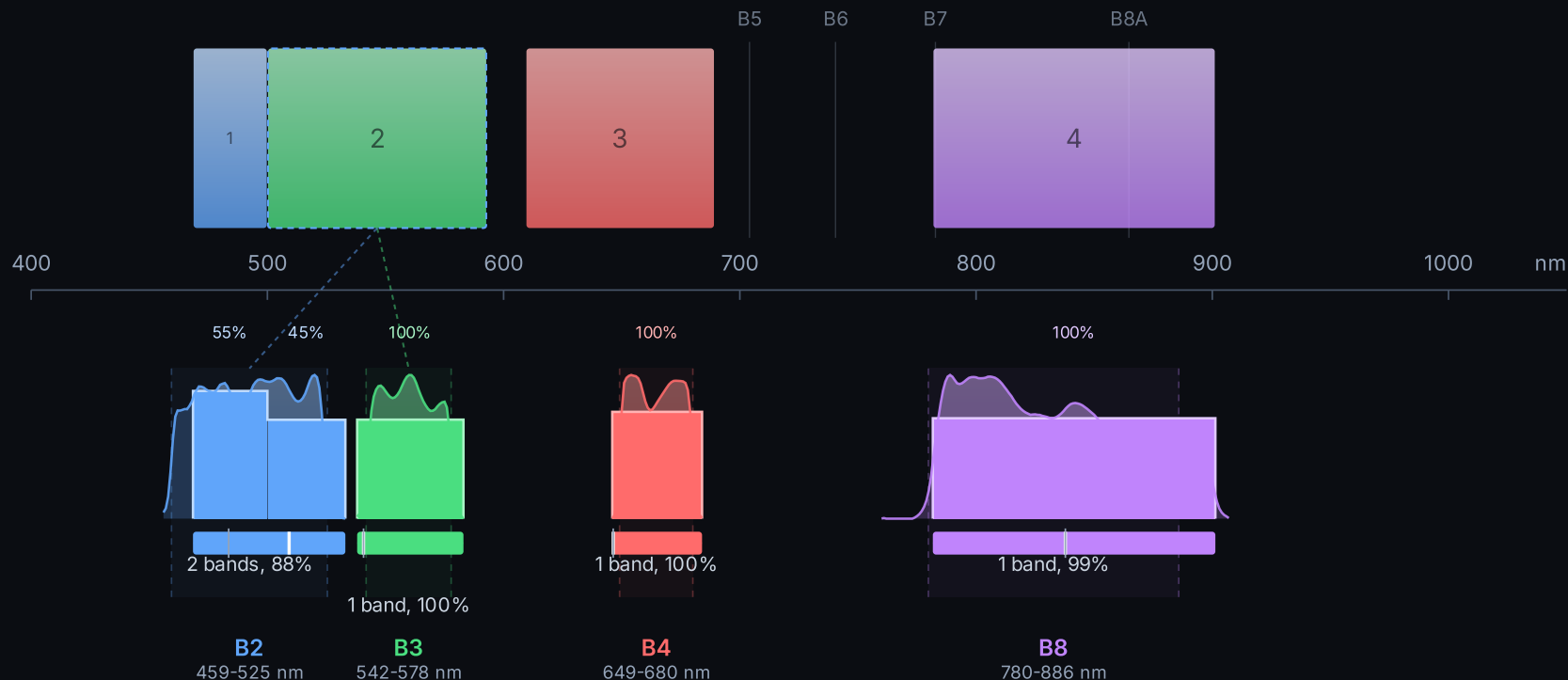
Mode 4 : 18 bands, ~11 m



Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Band 2 is 93 nm wide: it takes 45% of the blue mix next to band 1 and all of the green. That double duty is what pulls the blue centroid about 26 nm toward the green.

Plain mean vs SRF weights

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	7 → 8	3 → 5	3 → 4	10 → 14
2	2 → 3	2 → 2	3 → 3	2 → 2
3	1 → 2	2 → 2	2 → 2	2 → 3
4	1 → 1	1 → 1	1 → 2	2 → 3
5	1 → 2	2 → 2	2 → 3	4 → 7
20	1 → 2	1 → 1	1 → 1	1 → 1

CHRIS bands entering each pseudo-band. Left number is the plain mean set, right number the SRF set. Mode 5 band 22 comes back on its own, about a fifth of the B8 mix, and in mode 20 the SRF hands band 2 a share of the blue.

METHOD 3 OF 3

Convolution with the CHRIS response

which every CHRIS band whose modeled response touches the S2A curve takes part

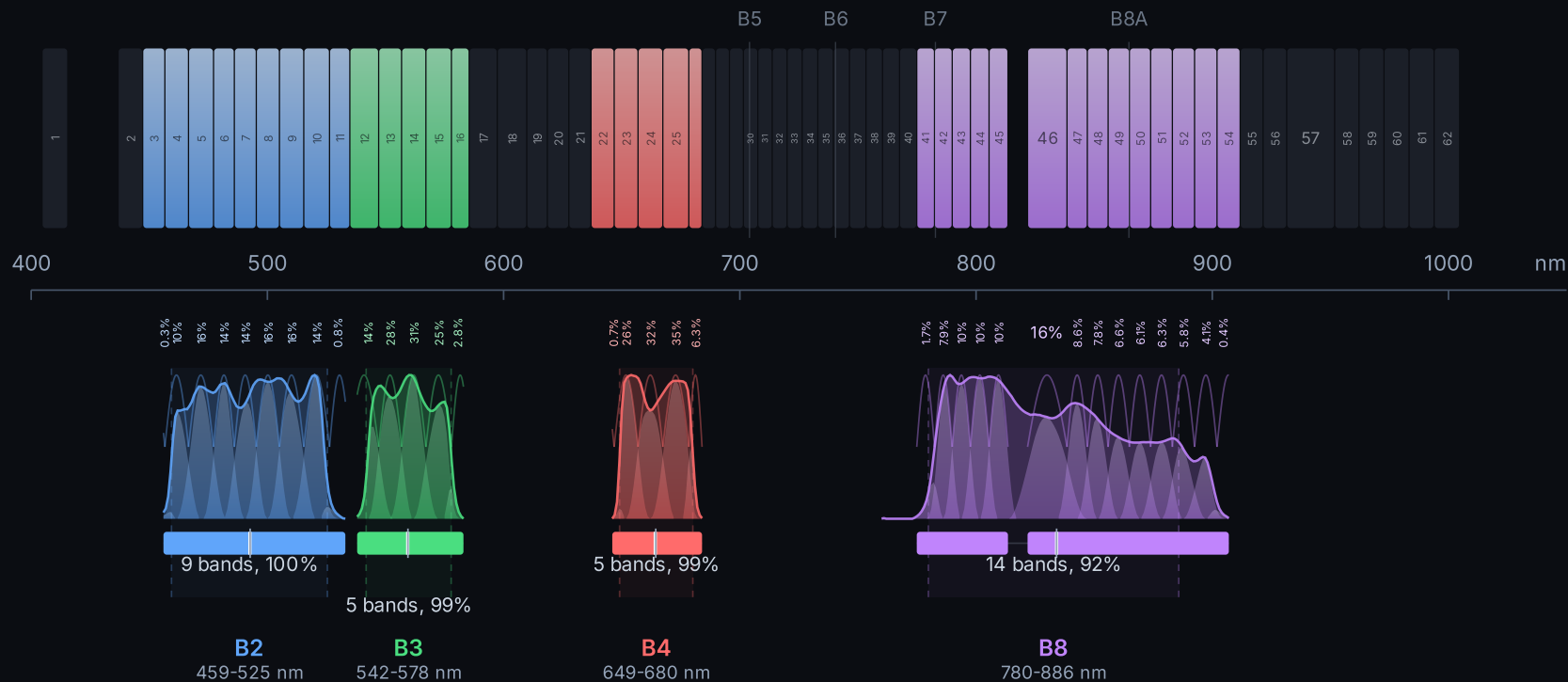
how w_i is the integral of the product of the two response curves

then CHRIS has no measured response, so R_i is a raised cosine fitted to WILow, WIMid and WIHigh, with a Gaussian of FWHM = BWidth as the control (sigma = BWidth / 2.3548)

$$\hat{L}_g = \frac{\sum_i w_i L_i}{\sum_i w_i}, \quad w_i = \int R_g(\lambda) R_i(\lambda) d\lambda$$

The CHRIS response is modeled, not measured: no CHRIS SRF exists in the data, so the kernels are fitted to WILow, WIMid and WIHigh. In the lanes: the raised-cosine outline is the modeled response and the solid area is its product with the S2A curve, the band weight. The percentages above the lanes are each band's share of the group weight.

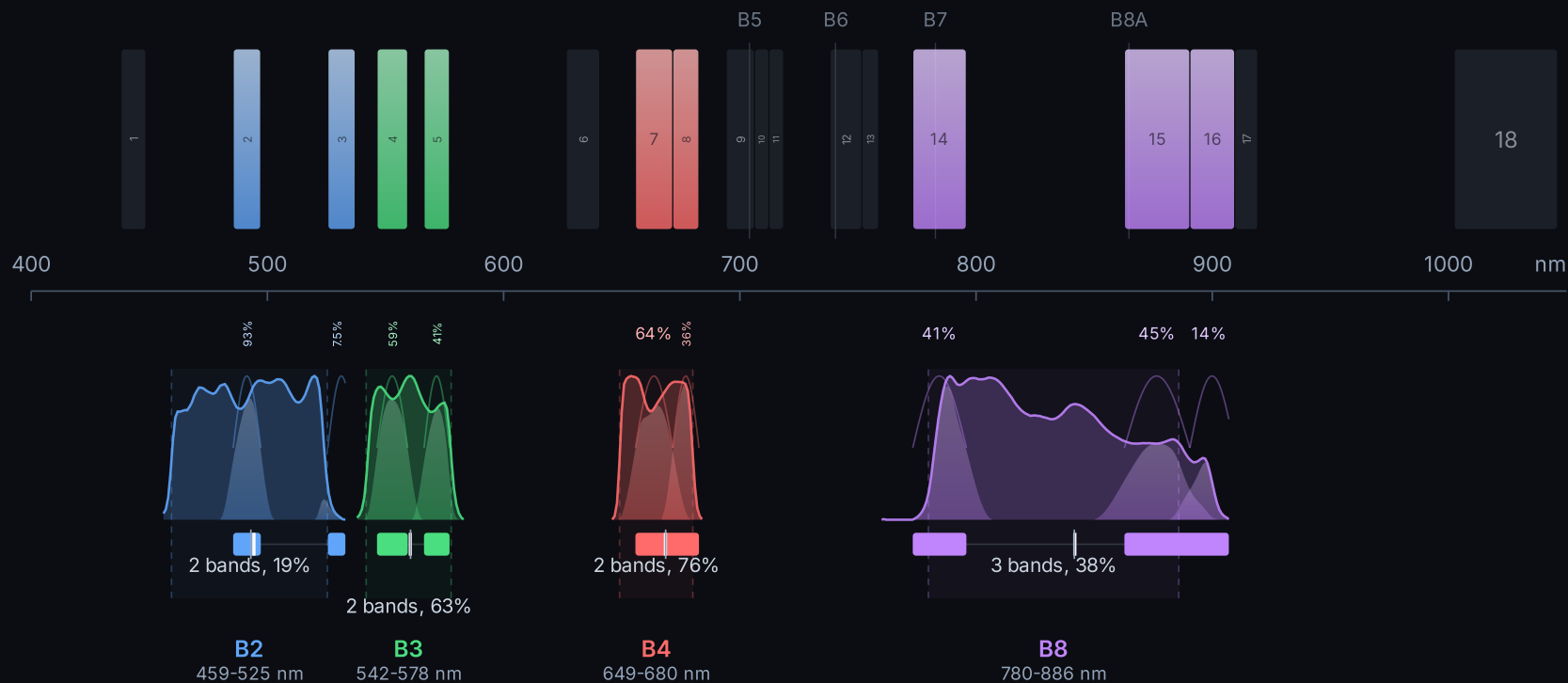
Mode 1 : 62 bands, ~34 m



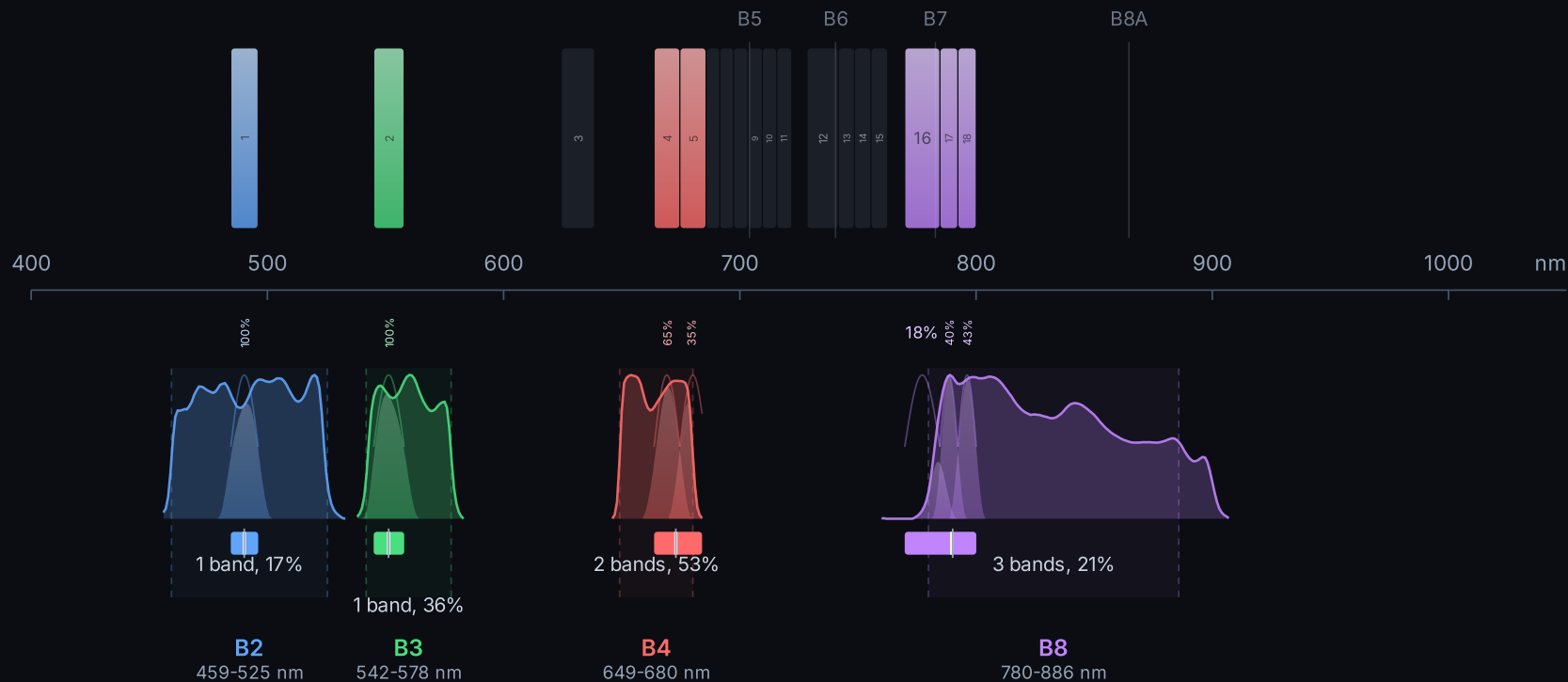
Mode 2 : 18 bands, ~17 m



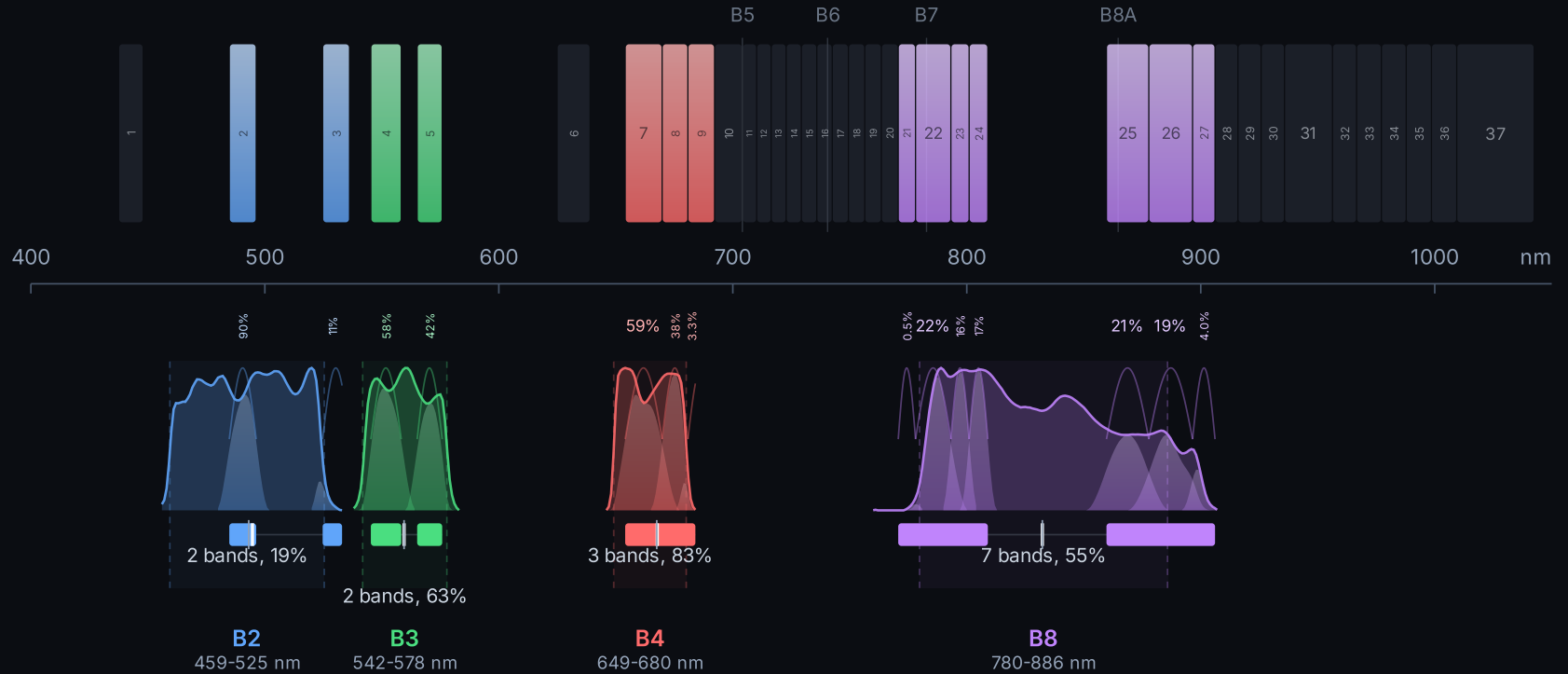
Mode 3 : 18 bands, ~13 m



Mode 4 : 18 bands, ~11 m



Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Raised cosine vs Gaussian

Metric	Raised cosine	Gaussian
Maximum modeled coverage	0.998 (Mode 1 B2)	1.061 (Mode 1 B2)
Bands shared across more than one target	1	5
Largest centroid shift against method 2 (nm)	1.6 (Mode 5 B2)	5.5 (Mode 20 B3)
Median RMSE of the emulated curve	0.361	0.353

Three-way comparison: band centroids

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	490.9 → 492.7 → 492.6	562.1 → 559.5 → 559.4	662.2 → 664.4 → 664.2	832.3 → 834.1 → 834.1
2	500.2 → 501.9 → 502.5	567.9 → 566.3 → 566.3	667.4 → 664.6 → 664.7	827.8 → 834.6 → 834.5
3	491.3 → 493.0 → 494.3	562.2 → 560.6 → 560.5	670.4 → 668.6 → 668.5	830.4 → 841.6 → 841.8
4	490.2 → 490.2 → 490.2	551.3 → 551.3 → 551.3	669.1 → 672.9 → 672.9	792.4 → 790.1 → 789.7
5	490.5 → 493.1 → 494.7	561.0 → 559.5 → 559.5	668.5 → 667.5 → 667.7	839.5 → 832.3 → 832.3
20	483.6 → 509.2 → 509.5	540.7 → 540.7 → 540.7	646.4 → 646.4 → 646.4	837.8 → 837.8 → 837.8

Cells read plain mean → SRF weights → convolution, in nm of centroid. The kernel barely moves the centroid except where coverage is low (Mode 3 and Mode 5 B2) and in Mode 20.

Final step: pruning

Tail contributions below 2% of the mix are dropped and the remaining weights are rescaled to 100%. It removes noise without touching the ranking.

Method	Entries	Dropped	Bands (weight before pruning)
SRF (box)	76	5	Mode 1 B2 b11 0.5%, Mode 1 B8 b41 1.5%, Mode 1 B8 b54 0.3%, Mode 5 B4 b9 1.2%, Mode 5 B8 b21 0.3%
Raised cosine	79	7	Mode 1 B2 b3 0.3%, Mode 1 B2 b11 0.8%, Mode 1 B4 b22 0.7%, Mode 1 B8 b41 1.7%, Mode 1 B8 b54 0.4%, Mode 2 B3 b8 0.6%, Mode 5 B8 b21 0.5%

Example, Mode 1 B8: the 14 entries lose their two smallest tails (1.5% and 0.3% in SRF, 1.7% and 0.4% in raised cosine); the 12 that stay shift up, from 4.2-15.4% to 4.3-15.7% (SRF) and from 4.1-15.6% to 4.2-15.9% (raised cosine). The order does not change.

The released weight tables apply this pruning: drop under 2%, then renormalize the rest.

Validation with real Sentinel-2

The four weight tables are only numbers until they are checked against real data. This section tests them on real S2 L1C (TOA reflectance) over the same targets.

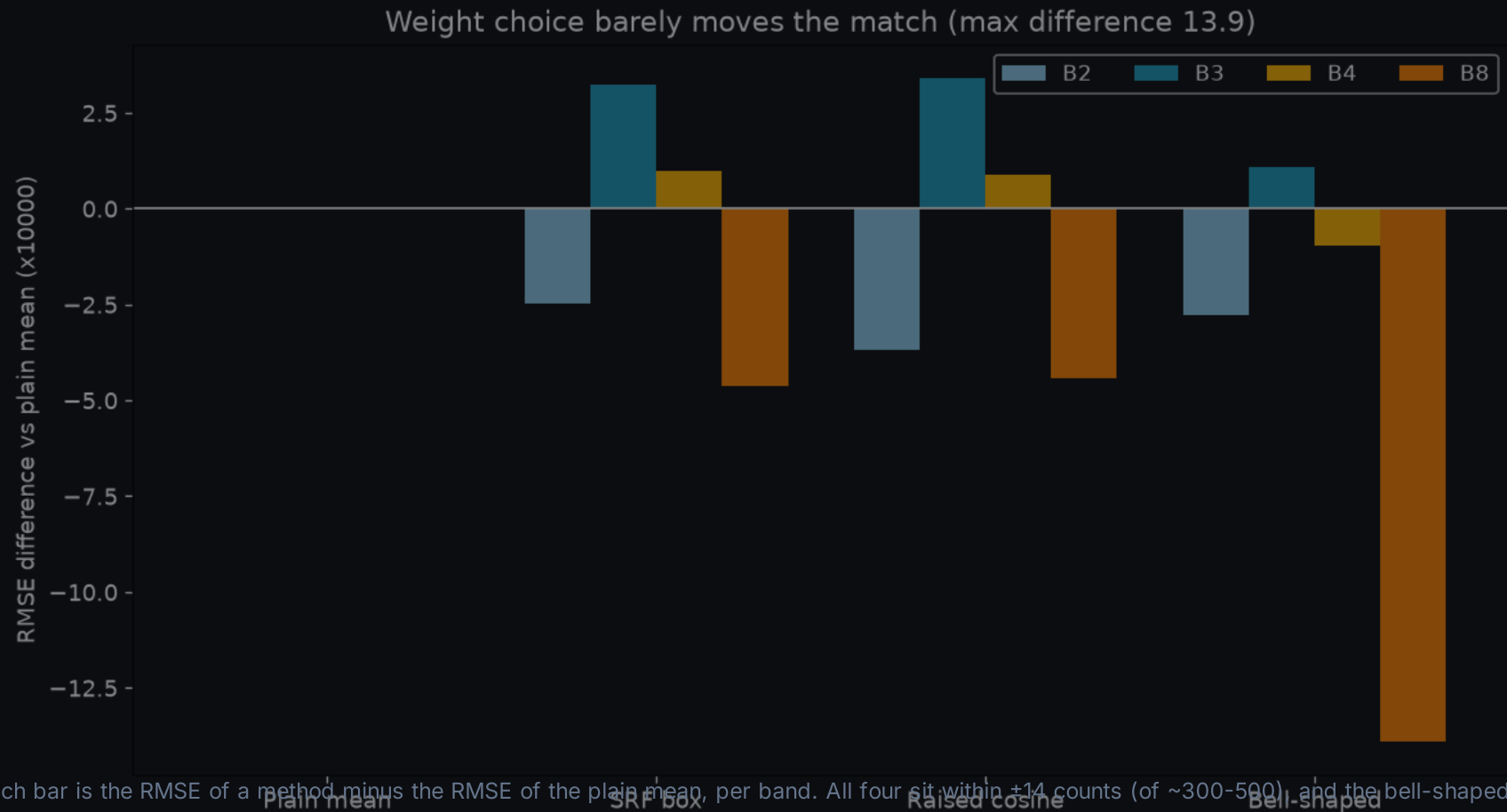
- 7 aligned CHRIS/S2 pairs across 4 modes (1, 3, 4, 5). Mode 2 is excluded: its L1A products since 2015 are radiometrically uncalibrated (gain flagged -1, RCI two to three orders of magnitude below spec). Mode 20 (mode 6 here) is DN-only, with no comparable TOA, so it sits out too.
- Pair rule: S2 as close in time as possible (1-2 days in 6 of 7), S2 clear ($s2cloudless \geq 95 \%$), CHRIS cloud $\leq 7 \%$ with the real cloud model.
- CHRIS pixels untouched on their native grid; S2 aggregated by area onto that grid, only in the intersection.

The pairs

Mode	Site	CHRIS date	S2 date	Platform	Δ days	S2 clear	NCC
1	Marseille, France	2019-09-11	2019-09-12	S2A	1	100 %	0.97
1	Nevada, USA	2021-12-17	2021-12-16	S2A	1	100 %	0.91
3	Madrid, Spain	2019-02-15	2019-02-16	S2B	1	99 %	0.62
3	Bangalore, India	2021-03-26	2021-03-24	S2A	2	97 %	0.82
4	Veneto, Italy	2020-07-10	2020-07-12	S2B	2	100 %	0.82
5	Gobi, China	2018-08-14	2018-08-13	S2A	1	99 %	0.85
5	Godavari delta, India	2016-04-17	2016-04-26	S2A	9	95 %	0.84

Alignment: initial pseudo-geotransform, coarse search (rotation, scale, translation), then ECC and LightGlue/LoFTR refinement. Peaks verified by a shift test.

Which weight set emulates S2 best?



Each bar is the RMSE of a method minus the RMSE of the plain mean, per band. All four sit within +14 counts (of ~300-500), and the bell-shaped kernel, the control in the weight section, wins by a hair. The choice of weights is not what limits the match.

Where the error lives

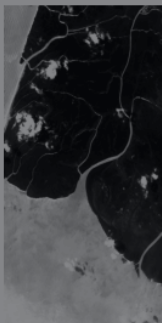
Band	Mean RMSE	Bias (CHRIS – S2)	Correlation
B2 blue	330	+180	0.65
B3 green	321	+135	0.68
B4 red	380	+70	0.71
B8 NIR	490	–12	0.75

Numbers are averaged over the four weight methods and the seven pairs, in reflectance x10000 (so 330 = 0.033). The visible bands carry a systematic positive bias (CHRIS reads brighter than S2); the NIR is nearly unbiased. That points at radiometry and atmosphere, not at the weights.

Same fields, two sensors

Best pair: Godavari delta, India (mode 5, S2 2016-04-26, 9 day gap)

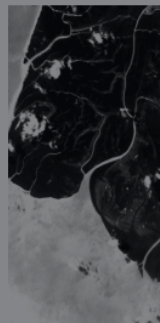
S2 B2



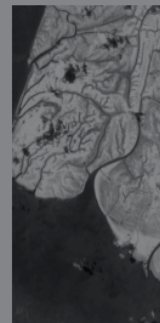
S2 B3



S2 B4



S2 B8



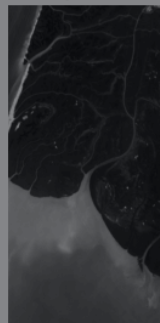
CHRIS pseudo B2



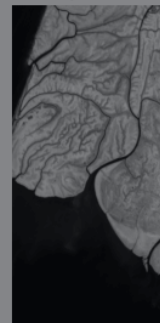
CHRIS pseudo B3



CHRIS pseudo B4



CHRIS pseudo B8



The best pair of the set: mode 5, 9 days apart, the only one that beats the 1-2 day pairs ($r = 0.92$). Same mask and same stretch per column in both rows; the pseudo-bands reproduce the field structure with the bell-shaped weights.

The spread tells the other half

Pair	Δ days	Mean correlation	Mean RMSE
Mode 5, Godavari delta	9	0.92	307
Mode 1, Marseille	1	0.91	248
Mode 1, Nevada	1	0.87	316
Mode 3, Bangalore	2	0.69	246
Mode 5, Gobi	1	0.57	545
Mode 3, Madrid	1	0.56	389
Mode 4, Veneto	2	0.36	601

Correlation runs from 0.36 to 0.92 and RMSE from 246 to 601. Time gap alone does not explain it: the best pair is 9 days apart and the worst 2. The hardest pairs are the ones with mixed land-water surfaces and thin residual cloud.

What this says

- The four weight tables are interchangeable at this level of agreement: the mapping is not the bottleneck.
- What limits the match is radiometry (visible bias), atmosphere and the time gap between acquisitions: things the weights cannot fix.
- Therefore the training mapping stays as published, and the weight table choice is documented as a second-order decision.

Next: rebuild CHRIS on the S2 grid in float (nodata NaN, explicit validity), and compare the pseudo-bands band by band with the released weights on all pairs.